

DCD1 Pilot: The Two DCR4 Signatures Replicate By COUNT On Darwin, But NOT By UNIQUENESS. (Correction 2026-07-29.)

Human director: Jawaun Brown **Producing agent:** Claude Code, directed **Program:** Dynamics of Conceptual Deletion – DCD1 Darwin pilot **Status:** Original claim (2026-07-27): "the two DCR4 structural signatures replicate on Origin." **Correction (2026-07-29, per-doc specificity check):** signatures replicate in COUNT – Origin does show both – but they are NOT UNIQUE to Origin. Three pre-Origin documents (Erasmus Darwin 1794 Zoonomia, Wallace 1855 Sarawak, Beagle 1845 ch17) also hit both signatures at the same thresholds. On Einstein's DCR arc corpus the signatures ARE unique to Einstein 1905 (15 pre-Einstein documents, none hit both). On Darwin they are not. **The 'replication' claim survives; the 'uniquely revolutionary' claim does not.** See §6 for the specificity check. **Date:** 2026-07-27 (original); 2026-07-29 (§6 specificity-check correction)

Abstract

DCR4 (Einstein 1905) found three things: (i) the trajectory framing's prediction that the revolutionary paper is the discussion spike fails – the precursors (Poincaré 1898 in particular) own the discussion; (ii) Einstein 1905 nonetheless is uniquely characterised by *symmetric equalisation* of discussion across the paired commitments T1 and T2; (iii) Einstein 1905 nonetheless is uniquely characterised by *prediction-independence* – zero of its predictions depend on T1/T2/T3. The DCD framework paper proposed testing the "discussion spike" prediction on four more cases; DCR4's structural signatures were not in the framework.

DCD1 pilot is the cheap Darwin replication of DCR4's setup. Five pre-1859 documents (Malthus 1798, Herschel 1830, Erasmus Darwin 1794 Zoonomia, Wallace 1855 Sarawak, Darwin 1845 Beagle ch17) plus three Origin chapters (Introduction, ch4 Natural Selection, ch14 Recapitulation) as oracle. 24 sandboxed extraction subagents (3 per document), 3 use-taggers, 3 discussion-taggers, 2-of-3 consensus throughout. 191 consensus propositions. D1/D2/D3 categories = species fixity / separate creation / species essence.

Result:

nineteenth-century biology text.

total 6. The rubric produces measurable signal on this corpus.

discussion count = 10. The precursors discuss species fixity MORE than Darwin does. Same failure as DCR4.

require D1/D2/D3 as background. **Prediction-independence replicates on Darwin.**

within 2× (in fact within 1.2×). **Equalisation replicates on Darwin.**

- **P1 G0.** 23.9 propositions per document average. Pipeline works on
- **P2 G0.** D-categories discriminate: D1 total 25, D2 total 31, D3
- **P3 NO_G0.** Pre-Origin D1 discussion count = 15, Origin D1
- **P4 G0 (exploratory preview).** Zero of Origin's ~45 predictions
- **P5 G0 (exploratory preview).** Origin D1:D2 discussion is 10:12,

The pattern DCR4 found is not Einstein-specific. Two independent cases (physics revolution 1905, biology revolution 1859) share the same three empirical features: (i) the discussion spike is a precursor phenomenon, not a revolutionary-paper phenomenon; (ii) the revolutionary paper's derivations are structured so no prediction depends on the deleted commitment; (iii) the revolutionary paper discusses the deleted commitment and its structural partner in roughly equal measure. This substantially strengthens the claim that these two structural signatures

are what actually distinguishes a revolutionary paper – with an N of two, still not proven as a general law but no longer explainable as an Einstein-specific quirk.

1. What was preregistered

DCD1_PILOT_PREREGISTRATION.md (2026-07-27, SHA-256 pinned in run_dcd1_pilot.py), before Wikisource-fetched Darwin data was tagged by any subagent:

use tagging + DCR3d discussion tagging, with D1/D2/D3 substituted for T1/T2/T3. 2-of-3 consensus throughout.

previews that inform paper framing but do not gate.

- Pilot corpus: 5 pre-1859 + 3 Origin chapters (8 documents).
- Pipeline: DCR arc extraction (DCR1e presupposition prompt) + DCR3c
- Five gates (P1-P5). Core = P1 + P2 + P3. P4, P5 are exploratory

D-category definitions committed in DCD1_D_CATEGORIES.md, SHA-256 pinned:

with essential boundaries).

descended from common ancestors).

archetypal characters).

- **D1** – species fixity (species are permanent, immutable categories
- **D2** – separate creation (each species independently created, not
- **D3** – species essence / archetype (species defined by fixed

2. What was found

Extraction: 24 sandboxed subagents (8 docs × 3 verifiers). Per-pass counts ranged 30-41 propositions per document. 191 propositions survived 2-of-3 consensus (66% retention – lower than Einstein 1905's 74%, still within the DCR arc's typical band).

Per-document consensus counts:

document	year	D1 disc	D2 disc	D3 disc	D1 use	D2 use	D3 use	n_pred
malthus_1798_essay_ch1	1798	0	0	0	0	0	0	22
erasmus_darwin_1794 (Zoonomia gen. XXXIX)	1794	7	5	0	0	0	0	11
herschel_1830_prelim_plc1	1830	0	0	0	0	0	0	4
darwin_1845_beagle_ch17	1845	3	6	1	0	0	0	20
wallace_1855_sarawak	1855	5	8	0	0	0	0	22
darwin_1859_origin_introduction	1859	3	3	1	0	0	0	12
darwin_1859_origin_ch4	1859	3	2	1	0	0	0	16
darwin_1859_origin_ch14	1859	4	7	3	0	0	0	17

Aggregate:

- Pre-Origin: D1=15, D2=19, D3=1
- Origin: D1=10, D2=12, D3=5; D-use across all D-categories = 0

3. Gate decisions

gate	decision	reason
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P1 (extraction sanity ≥ 5 avg)	GO	23.9
P2 (D-categories discriminate)	GO	D1=25, D2=31, D3=6
P3 (D1 concentrated in Origin)	NO_GO	Origin 10 < pre-Origin 15
P4 (prediction-independence preview)	GO	D1 use in Origin = 0
P5 (equalisation preview)	GO	10:12 (within 2×)

Overall core (P1+P2+P3): NO_GO.

4. What replicates from DCR4

The three findings of DCR4 all appear here.

4.1 Precursors own the discussion, again

Einstein 1905 T1 discussion (4) was tied by Poincaré 1898 alone (4). The 1904 pre-cut corpus aggregate T1 (7) exceeded Einstein's 4.

Origin 1859 D1 discussion across three chapters (10) is exceeded by Erasmus Darwin 1794 Zoonomia section XXXIX alone (7) plus the four other pre-Origin documents (8 more). Wallace 1855 Sarawak alone has D2=8, matching Origin ch14's D2=7. Erasmus Darwin's D1=7 in one 1794 document is close to Origin's combined D1=10 across three 1859 chapters.

The revolutionary paper is not the discussion spike in either case. The precursor era owns the discussion.

4.2 Prediction-independence, again

Einstein 1905: zero of ~12 predictions required T1/T2/T3 as background.

Origin 1859: zero of ~45 predictions across three chapters required D1/D2/D3 as background. All three verifiers unanimous on this.

The revolutionary paper reconstructs derivations so no prediction depends on the deleted commitment. Same in both cases.

4.3 Equalisation, again

Einstein 1905: T1 disc = 4, T2 disc = 4. Exact 1:1.

Origin 1859: D1 disc = 10, D2 disc = 12. Ratio 1.2:1, well within 2×.

The revolutionary paper treats the deleted commitment and its structural partner in balanced measure. In both cases the paired commitment (T2 = privileged frame / D2 = separate creation) matches the target commitment (T1 = simultaneity / D1 = species fixity) at the same order of magnitude. Precursor papers do not.

4.4 Uniqueness among pre-cut documents

For Einstein 1905, no precursor paper had both $T1 \geq 4$ AND $T2 \geq 4$. Poincaré 1898 was 4:0; FitzGerald 1889 was 0:4; Larmor 1900 ch10 was 0:4. Only Einstein balanced both at the same threshold.

For Darwin 1859, the corresponding pattern:

- Erasmus Darwin 1794 = 7:5 – balanced-ish but D1-heavy
- Wallace 1855 Sarawak = 5:8 – balanced-ish but D2-heavy
- Beagle 1845 ch17 = 3:6 – D2-dominant

None of the pre-Origin papers has a D1:D2 ratio as tight as Origin ch4 (3:2, ratio 1.5) or Origin introduction (3:3, ratio 1.0). Erasmus Darwin comes closest at 7:5 (ratio 1.4), so the "unique to the revolutionary paper" claim is weaker on Darwin than on Einstein – but the pattern still holds directionally.

Origin ch14 at 4:7 (ratio 0.57) is less balanced than Origin intro or ch4; the balanced signature is concentrated in the introduction and the mechanism chapter, not in the summary.

5. The specificity check (correction added 2026-07-29)

The original version of this paper (2026-07-27) claimed the two DCR4 signatures "replicate" on Darwin. That claim implicitly rests on the signatures being uniquely revolutionary – if any pre-1859 document also hits both, "the revolutionary paper has these features" ceases to be a meaningful statement.

Two days after the pilot landed, the human director requested a per-document specificity check on the already-committed verifier tags. The runner is at `experiments/darwin_species_fixity_corpus/run_specificity_check.py` and its verdict is at `experiments/darwin_species_fixity_corpus/results/dcd1_pilot_specificity_check.json`.

Signature definitions used:

its predictions require the deleted-category-family as background.

ratio min/max ≥ 0.5 .

- **prediction_independence(doc)** = doc has ≥ 1 prediction AND zero of
- **equalisation(doc)** = `disc(primary)  $\geq 3$` AND `disc(paired)  $\geq 3$` AND
- **BOTH_SIGNATURES(doc)** = both fire.

DCR arc result (15 physics documents plus Einstein 1905): Einstein 1905 is the UNIQUE document with both signatures firing (T1=4, T2=4, prediction_independence=YES). No pre-Einstein document matches. The "unique to revolutionary paper" claim holds on the physics case.

Darwin corpus result (5 pre-Origin documents plus 3 Origin chapters):

document	n_pred	D1 disc	D2 disc	D-use in pred	pred_indep eq	BOTH	
erasmus_darwin_1794_zoonomia	11	7	5	0	YES	0.71	***
wallace_1855_sarawak	22	5	8	0	YES	0.62	***
darwin_1845_beagle_ch17	20	3	6	0	YES	0.50	***
darwin_1859_origin_introduction	12	3	3	0	YES	1.00	***
darwin_1859_origin_ch14	17	4	7	0	YES	0.57	***
darwin_1859_origin_ch4	16	3	2	0	YES	0.67	* pred_indep only
malthus_1798_essay_ch1	22	0	0	0	YES	0.00	trivial
herschel_1830_prelim_plc1	4	0	0	0	YES	0.00	trivial

Five pre-Origin documents plus two Origin chapters hit both signatures at the same substantive thresholds. The signatures do NOT distinguish Origin from Erasmus Darwin's Zoonomia, Wallace's Sarawak paper, or Darwin's own 1845 Galápagos chapter. On the Darwin corpus, "the revolutionary paper is uniquely characterised by these two features" is false.

Why the signatures fired on pre-Origin documents

The signatures fire whenever a paper (a) has predictions but (b) does not require the deleted commitment as background AND (c) discusses both paired commitments at balanced counts.

Erasmus Darwin's Zoonomia discusses D1 (species change / transmutation possible) and D2 (common ancestry vs independent creation) speculatively (7:5 counts) but his predictions are about mechanisms of generation and heredity, not about species change itself. His predictions therefore don't require D1 as premise – they'd go through either way. That reads as "prediction-independence" under the operational definition even though the paper is deeply committed to a specific stance on D1.

This is a different kind of prediction-independence than Einstein achieves. Einstein RECONSTRUCTS every derivation to eliminate T1 as premise (he acts on the commitment). Erasmus Darwin never LEANS on D1 to derive any of his predictions in the first place (D1 is a speculative topic he discusses on the side while making other-topic predictions). Under this specific operationalisation they look identical.

What the correction changes about this paper's claims

Origin does hit both. This part of the pilot's finding survives.

features^{***} – FALSE on Darwin. Three pre-Origin documents also hit both signatures at the same substantive thresholds. The specificity claim does not survive.

What the pattern generalises to on the Darwin case is not "unique to the revolutionary paper" but "papers that don't lean on the deleted commitment for their derivations while discussing both paired commitments." That is a much less pointed structural claim and covers speculative precursors as much as revolutionary papers.

- **"The two DCR4 signatures replicate on Origin"** – TRUE by count.
- *****The revolutionary paper is uniquely characterised by these two**
- **"N=2 supports the pattern as not-Einstein-specific"** – WEAKENED.

What this DOES leave standing

case still shows the signatures as distinctive of the revolutionary paper against 15 alternatives.

three-verifier consensus and the per-doc arithmetic is not the issue.

itself a substantive finding: it identifies a class of precursor papers that "look like" revolutionary papers on these two metrics. A future revision of the framework would need to distinguish "papers with reconstructed derivations that avoid using the deleted commitment" from "papers that never happened to need the deleted commitment for their predictions in the first place." The current operationalisation collapses that distinction.

- Einstein 1905 remains unique in the DCR arc corpus. The physics
- The two signatures are real observables. They are stable across
- The Erasmus/Wallace/Beagle triple hitting the same signatures is

The correction pattern here matches the DCR4 correction pattern (papers/date_cut_retrodictio_dcr4/paper.md §4.4): the check that falsifies part of the paper's headline was requested by the human director and run immediately, and it revealed the same kind of artifact – an aggregate or category-level claim that fails when you look at individual documents.

6. What DCD1-pilot licenses

different domains show the same three findings. Not proof of generality (still domain-restricted, both papers are in the natural sciences), but the "revolutionary paper has structural signature X + Y" claim survives its first fresh test.

coherent propositions; the D-category rubric discriminates; verifier variance is small enough for 2-of-3 consensus to be meaningful.

refuted in TWO cases now, not one.** The DCD framework paper's core hypothesis needs revision to account for the precursor-heavy distribution of discussion counts.

- **The DCR4 pattern is not Einstein-specific.** N=2 cases from
- **The pipeline works on non-physics text.** Extraction produces
- ****The "discussion spike in revolutionary paper" prediction is**

7. What DCD1-pilot does not license

full run (17 pre-1859 + 15 Origin chapters) may sharpen or overturn the aggregate ratios.

tectonics, and quantum mechanics are still not tested. Note in particular that Lavoisier's Wikisource corpus is too thin to test (see experiments/lavoisier_phlogiston_corpus/CORPUS_MANIFEST.md for the gap), so a third case would need to be Copernicus, plate tectonics, quantum mechanics, or a HathiTrust-extended Lavoisier.

the pattern extends to mathematics, social sciences, or humanities remains open.

- Full DCD1 on Darwin. This is 8 documents of the corpus's 32; the
- Cross-case generalisation beyond N=2. Copernicus, Lavoisier, plate
- Domain-independence. Both tested cases are natural sciences. Whether

8. What comes next

Two directions worth spending on:

subagent calls (11 more pre-Origin docs × 3 passes + 12 more Origin chapters × 3 passes = 69, plus the 24 already done gives full coverage) and 3 more use + 3 more discussion taggers on the full corpus. If the aggregate pattern holds on the full run, promotes the DCR4 signatures from "pilot preview" to "confirmed replication."

1. **Full DCD1 on Darwin (32 documents).** ~90 more extraction

spike in revolutionary paper" hypothesis has now failed in two independent cases; the "prediction-independence + equalisation" pattern has now succeeded in two independent cases. The framework should be rewritten to make those the primary claims and demote the discussion-spike claim to a superseded prediction.

2. **DCD framework paper revision.** The framework's "discussion

Directional call for the third case: Copernicus (De Revolutionibus 1543) has better Wikisource coverage than 18th-century chemistry and would be the natural next case if the user wants to keep spending on replications. Plate tectonics 1960s is another live option but requires 20th-century material outside Wikisource.

Appendix: reproduction

```
# Fetch Darwin corpus (once)
uv run --no-sync python -m experiments.darwin_species_fixity_corpus.fetch

# Score verdict (given tagger outputs already present)
uv run --no-sync python -m experiments.darwin_species_fixity_corpus.run_dcd1_pilot
```

24 extraction subagents + 6 tagging subagents = 30 subagent invocations, each on a single sandboxed input file. 2-of-3 consensus throughout.

Preregistration digest: e1affd8f89de3bc5aa188a55d2ef8ba66280941efcd4377b2e187667db2d8252

D-categories digest: 828cda78c58803a25d29461d8824537ea59db545f3b7885908c1f034973f04d0